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CSA1506

Cloud Computing

SHORT ANSWER

Q Define cloud Computing and explain any four characteristic that make it different from traditional on premise computing.

sol:- cloud computing is the on demand delivery of computing services - including servers, storage, databases, networking, and software - over the internet with pay-as-you-go pricing

Characteristics vs on premise:-

on demand self-service:- Users can provision computing capabilities automatically without requiring human interaction with the service provider.

Elasticity and scalability: Resource can be scaled up or down rapidly based on demand: on premise requires manual hardware upgrades.

Pay-as-you-go pricing: you pay only for what you use (OpEx), where as on-premise requires significant upfront capital investment.

Broad Network Access: Services are available over the network and accessed through standard mechanisms.

② Trace evaluation of Cloud Computing

Hardware Evolution

Large mainframe computers were used in organizations.

Personal Computers: Affordable PCs made computing accessible to individuals.

client-server Architecture: clients requested services from centralized servers.

virtualization - Multiple virtual machines can run on a single physical server, improving efficiency.

Internet Evolution - high-speed Internet enabled remote access to applications and data.

Web Applications - Software became available through web browsers.

Cloud Computing - Providers like AWS, Microsoft Azure, and Google Cloud began offering scalable services over the Internet.

③ Explain the role of Server virtualization

Role of Server virtualization

- Divides one physical server into multiple virtual machines.
- Improves hardware utilization.
- Reduces infrastructure cost.
- Enables quick deployment and easy management of resources.

Virtualization	Cloud Computing
→ Technology that creates virtual computers virtual machines → Focuses on efficient use of hardware → Runs on a physical server <u>Ex:-</u> VMware, Hyper-V	→ services model that delivers computing resources over the internet → Focuses on providing services to users → Uses virtualization to provide scalable services. <u>Ex:-</u> AWS, Microsoft Azure.

④ Discusses the role of data centers in delivering cloud services.

sol:- Data centers are facilities that contain servers, storage devices, networking equipment, and cooling systems used to provide cloud service.

Roles

- Store and process user data
- Host cloud application and website

- Provide secure backup and disaster recovery.
- Ensure high availability and reliability.
- Enable virtualization and resource sharing.
- Deliver cloud services to users world wide through the Internet.

⑤ Differentiation Iaas, Paas, Saas & Xaas with one suitable real-world example.

Service Model	Description	Example
Iaas (Infrastructure as a Service)	provides virtual servers storage and networking	Amazon EC2 (AWS)
Paas (Platform as a service)	provides a platform to develop and deploy applications	Google App (Engine)
Saas (Software as a service)	Delivers software applications over the Internet	Google Docs
Xaas (Anything)	provide various IT services through the cloud.	Microsoft 365